

TANAY TAMMINENI

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Professional Summary

Innovative AI Software Engineer with expertise in building autonomous agentic workflows, production-grade RAG pipelines, and distributed LLM fine-tuning infrastructure. Proficient in architecting scalable Python backend systems, designing ETL pipelines, and orchestrating complex multi-stage coordination tasks via LangGraph and DeepSpeed across multi-GPU environments. Dedicated to stress-testing intelligent systems, optimizing real-time data performance, and deploying robust generative AI solutions on AWS, Azure, and GCP at scale.

Technical Toolkit

Languages/Frameworks: Python, SQL, Java, FastAPI, LangChain, LangGraph, REST APIs, PyTorch, TensorFlow, Scikit-learn
AI/ML Ops: RAGAS, LlamaIndex, QLoRA, DeepSpeed, vLLM, DPO, YOLOv8, OpenCV, MinHash LSH, OpenAI Whisper
Infrastructure/Data: AWS (S3, Bedrock, DynamoDB, Lambda), Azure, GCP/Vertex AI, Docker, Terraform, CI/CD
Storage/Monitoring: PostgreSQL, Qdrant, Elasticsearch, Neo4j, MongoDB, Redis, Databricks, PySpark, Prometheus, Grafana

Professional Experience

VoiceBotics AI

AI Systems Developer Intern

Remote
Apr 2025 — Present

- Developed a FastAPI WebSocket server using Python asyncio with a 60-second reconnect window and session management, achieving a 94% preflight success rate and ensuring persistent streaming across high-latency environments for real-time AI applications.
- Built an STT latency watchdog with 5s warn and 8s degrade thresholds, automating recovery after 5 consecutive degraded batches and maintaining mean transcription TTF under 2.5s across live Zoom and Teams sessions.
- Implemented 16 normalized error codes with platform-specific root cause mapping and JSONL instrumentation logging for every system event, generating automated beta metrics summaries tracking session counts, error breakdowns, and STT state transitions.
- Engineered the LiveWire Chrome extension using tabCapture API and FFmpeg stereo splitting for dual-channel audio capture in Zoom, Teams, and Google Meet, enabling real-time AI transcription with automated evidence pack generation per session.

Globalshala

Software Engineer Intern

Hyderabad, India
Jun 2022 — Dec 2022

- Managed Java and SQL data management systems for internal cloud-native Azure integrations, maintaining 99.9% uptime across enterprise data ingestion pipelines processing 10,000+ daily records.
- Built Azure Databricks ETL pipelines to process organizational data for Power BI dashboards, delivering real-time KPI insights to 50+ stakeholders and reducing report generation time by 70%.
- Optimized SQL schemas across distributed enterprise systems improving data retrieval speeds by 40%, supporting automated reporting and consistent data flow across high-throughput pipelines.

Education

Southeast Missouri State University

M.S. in Computer Science — GPA: 3.9/4.0

Cape Girardeau, MO
Jan 2024 — Dec 2025

Technical Projects

Production RAG System

Python, LangChain, LangGraph, Qdrant, Elasticsearch, Neo4j, FastAPI

- Built a hybrid retrieval system combining Qdrant dense search and Elasticsearch BM25 via Reciprocal Rank Fusion, achieving 23.7% retrieval faithfulness gain with context precision of 1.0 across 50 evaluated queries at 163.5ms mean latency.
- Implemented CRAG as a LangGraph state machine with automated query rewriting and web search fallback, triggering self-correction on 38% of queries and achieving 0.87 answer relevancy across the arXiv research corpus evaluation.
- Integrated a RAGAS evaluation dashboard monitoring faithfulness (0.43), context precision (1.0), and answer relevancy (0.87) in real time via FastAPI and React, leveraging Neo4j graph relationships to enhance contextual depth.

Distributed LLM Fine-Tuning Pipeline

Python, PyTorch, QLoRA, DeepSpeed, vLLM, Terraform, Docker

- Built a distributed pipeline across 66 files and 11,000+ lines using QLoRA and DeepSpeed ZeRO-3 to fine-tune Llama 3 8B, achieving 41.2% per-GPU memory reduction via MinHash LSH deduplication with 128 permutations across multi-GPU nodes.
- Deployed models with vLLM and AWQ quantization achieving 3.8x inference throughput improvement, with Prometheus and Grafana monitoring on Terraform-provisioned AWS infrastructure tracking real-time GPU utilization.
- Published findings in UniLLMOps (Zenodo DOI: 10.5281/zenodo.19582347) documenting end-to-end pipeline architecture covering QLoRA + DPO training, TIES/DARE/SLERP model merging, and vLLM deployment.

Real-Time Vehicle Detection

Python, YOLOv8, OpenCV, PyTorch, TensorFlow

- Developed a YOLOv8 object detection model processing 5,000+ video frames with 88% accuracy for automated vehicle counting and classification across multiple vehicle classes in real-time traffic surveillance.
- Optimized inference by 25% using TensorFlow model quantization and OpenCV image processing pipelines for real-time edge performance, enabling deployment on low-power hardware while maintaining high-speed frame analysis.
- Published research in CVR Journal Vol.24 (June 2023) and secured 3rd Prize at Project Expo 2K23 for innovation in intelligent transportation systems.

Publication & Awards

- Published:** “UniLLMOps: A Unified Framework for End-to-End LLM Production Systems” — Zenodo, DOI: 10.5281/zenodo.19582347 (Apr 2026).
- Published:** “Real-Time Video Based Vehicle Detection, Counting & Classification” — CVR Journal, June 2023 | **Award:** 3rd Prize, Project Expo 2K23.
- Certifications:** Prompt Engineering & Programming with OpenAI — Columbia Engineering (Mar 2026) | Python Essentials 1 — Cisco (Jul 2023) | Programming Fundamentals — Duke.